

Brendan Boyd

boyd.brendan@stonybrook.edu

biboyd.github.io

+1 (248) 697-7902

Education

SUNY Stony Brook University Physics & Astronomy Department <i>PhD in Physics, with Concentration in Astronomy</i>	2020-2026
Michigan State University College of Natural Science, Honors College <i>Bachelor of Science, Astrophysics</i> <i>Minors in Math and CMSE</i>	2016-2020

Research & Teaching Experience

Graduate Research Assistant – Stony Brook University Computational Astrophysics Dissertation Research. Using a low Mach reactive hydrodynamic code to study the convective Urca process in relation to type Ia supernovae. Multiple publications in academic journals.	2022-Present
Teaching Assistant – Stony Brook University	2020-2021
Undergraduate Research Assistant – Michigan State University Astrophysics Thesis Research. Developed SALSA Python package for analyzing galaxy simulations. Publication in the Journal for Open Source Software (JOSS).	2019-2020
Undergraduate Learning Assistant – Michigan State University	2019-2020

Computational Skills

Programming Skills:

Proficient in Python
Proficient in C++
Competent in FORTRAN
Competent in HPC workflows (e.g. SLURM, bash scripting)
Competent in OpenMP threading
Competent in MPI parallelism

Computational Fluid Modeling

Low Mach Flows
(Nuclear) Reactive Flows

Software Projects - Contributor/Developer

MAESTROeX - github.com/AMReX-Astro/MAESTROeX	2022-Present
pynucastro - github.com/pynucastro/pynucastro	2022-Present
initial models - github.com/AMReX-Astro/initial models	2022-Present
SALSA - github.com/biboyd/SALSA	2019-2021